

NEDA KEYHANVAR, PhD

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PROFESSIONAL SUMMARY

PhD-trained molecular and stem cell biologist with over 12 years of research experience, including 3 years of postdoctoral training at UCSF, specializing in functional genomics, CRISPR-based genome engineering, and pooled perturbation screening. Experienced in recombinant DNA vector construction, molecular cloning, lentiviral vector production and transduction, mammalian cell culture, RT-qPCR, flow cytometry/FACS, and single-cell genomics workflows. Designed and executed a 12,000-guide pooled CRISPRi screen across 150 million sorted cells and engineered CRISPR knock-in hiPSC reporter lines to investigate epigenetic regulation. Collaborative scientist with a proven record of developing scalable experimental platforms and translating complex molecular biology workflows into reproducible, high-impact discoveries supporting next-generation therapeutic development.

PROFESSIONAL EXPERIENCE

Senior Postdoctoral Researcher | UCSF – Martens Lab | San Francisco, CA **Feb 2025 – Present**

- Maintained experiment documentation to GLP standards using Benchling ELN, ensuring data integrity, managing inventory tracking for equipment, reagents, and specimens while supporting controlled substances compliance aligned with corporate QSR and ISO compliance frameworks.
- Built an end-to-end NGS-based multi-omics pipeline spanning primary cell isolation, multi-color FACS sorting, bulk RNA-seq, and scRNA-seq sample preparation, supporting multiple concurrent projects.
- Independently built the lab's first decellularized thymic ECM scaffold platform across 5 age timepoints, establishing a reproducible mechanobiology pipeline and opening an entirely new research program.
- Collaborated with a computational postdoc to integrate mechanobiology with single-cell multi-omics, expanding the lab's cross-disciplinary analytical capabilities.
- Performed multi-color immunophenotyping of thymic stromal and immune cell populations by flow cytometry to characterize the effect of YAP/TAZ loss on immune cell composition and T cell development.
- Engineered and characterized a cKO transgenic mouse model demonstrating YAP/TAZ ablation accelerates thymic involution (19 days vs. 8-9 months naturally), establishing a preclinical aging model; served as lab's sole *in vivo* specialist performing 20+ sterile survival surgeries across multiple transgenic lines.
- Mentored a research associate for 18 months through 5 graduate school interviews resulting in PhD program admission; supervised an undergraduate researcher and a rotating PhD student.
- Selected for oral presentation and poster at the Bakar Aging Research Institute (BARI) Symposium (2025).

Postdoctoral Researcher | UCSF – Vissers Lab | San Francisco, CA **Oct 2023 – Jan 2025**

- Independently engineered hiPSC reporter lines via CRISPR/Cas9 knock-in using a novel split-GFP strategy (GFP11 tag only), improving feasibility over prior approaches and enabling first-in-lab endogenous tagging of m6A RNA methylation machinery.
- Executed the lab's first pooled CRISPRi screen with a 12,000-gRNA library across 150M sorted cells, identifying upstream regulators of epitranscriptomic networks and performed hiPSC-to-NPC differentiation and hypoxia tests as functional validation of CRISPRi screen hits.
- Routinely performed Real-time PCR (RT-qPCR) for gene expression validation and target confirmation across multiple projects, including downstream validation of CRISPRi screen hits and stem cell differentiation assays.
- Independently established lentiviral vector production (HEK293T), transduction, and nucleofection workflows; applied FACS-based high-throughput screening for clone selection and characterization of engineered mammalian cell lines across multiple rounds of selection.
- Drafted comprehensive lab reports via Benchling ELN, experimental protocols, and technical conclusions utilized for Design History Files (DHF); presented routine data findings during multi-functional group meetings.
- Contributed experimental input to a PhD student's brain organoid model, resulting in co-authorship on a published preprint; mentored a research associate for 13 months supporting a successful graduate school application.

Research Scientist | Stem Cell and Regenerative Medicine Institute | Tabriz, Iran **Sep 2014 – Sep 2023**

- Independently engineered a thermoresponsive polymer scaffold directing adipose-derived stem cell differentiation into keratinocyte-like cell sheets, developed end-to-end from biomaterial design through functional validation.
- Built and maintained a sterile 4-cell-type primary biobank (MSCs, fibroblasts, keratinocytes, adipose-derived stromal cells) including cryopreservation protocols, supporting translational regenerative medicine programs.
- Published 8 peer-reviewed publications (3 first author); presented at 2 international conferences; served as invited peer reviewer for BioRxiv journal (6 manuscripts).

- Trained and mentored 15+ students in experimental design, data interpretation, and scientific communication; established as informal team lead and primary technical resource within the lab.

SKILLS

Molecular Biology & Genomics: CRISPR/Cas9 knock-in/KO, CRISPRi, pooled library screening, host cell engineering, expression vector design & cloning, Real-time PCR, western/southern blotting, ELISA, molecular cloning, bacterial & insect Expression Systems, lentiviral & baculoviral vector production, gene delivery in mammalian cells

Functional Assays & Imaging: Multi-colour Flow cytometry & FACS sorting (BD FACSAria Fusion, FACSDiscover S8, Attune NxT), FlowJo, immunophenotyping, functional assays, confocal microscopy, fluorescence imaging, FIJI/ImageJ.

NGS based Multi-Omics & Bioinformatics: Bulk RNA-seq, single-cell RNA-seq (scRNA-seq), spatial transcriptomics & CUT&Tag sample preparation, NCBI, BLAST, PDB, UniProt, ExPASy

Stem Cell & Cell Biology: Aseptic technique, hiPSC culture & differentiation (into NPCs, neurons), mesenchymal stem cell (MSC) culture & differentiation, primary cell isolation (mouse & human: fibroblasts, keratinocytes), immortalized cell lines (HEK293T, ARPE-19), immune tissue biology (thymus, T cell development, immune aging), cell sheet engineering, Tunable stiffness & patterned substrates (PDMS), immunohistochemistry (IHC), immunocytochemistry (ICC) & immunofluorescence (IF) staining

In Vivo Research (Mouse Models): Transgenic mouse models of disease & aging, colony management & maintenance (breeding/weaning/genotyping/health monitoring), barrier facility import, *In vivo* injections and animal dosing (intraperitoneal (IP), subcutaneous (SC), tail vein (IV), oral gavage (PO)), survival surgery (thoracotomy intrathymic injection), cardiac perfusion, necropsy, Euthanasia & Anesthesia, tissue fixation, processing, and embedding (OCT/FFPE), Cryo & microtome sectioning, IACUC certification (BRER I/II/III), IACUC protocol compliance, Biosafety & controlled Substances, GLP (Good Laboratory Practice)

Software & Documentation: Data Integrity Standards, Controlled substance & GLP aligned documentation, Benchling (ELN), GraphPad Prism, FlowJo, FIJI/ImageJ, BioRender, Google apps, Microsoft (Excel, Word, Outlook, PowerPoint), Adobe Creative Suite, SOP development, technical report writing, Slack, Zoom

Leadership & Professional Skills: Experimental design & optimization, protocol development & standardization, Scientific writing, peer-reviewed publication & technical report preparation, Data presentation & visualization, Cross-functional collaboration, team mentorship & training, analytical troubleshooting, project management, grant writing, Oral & written communication skills, detail-oriented, strategic planning, multi-tasking, time management

EDUCATION

PhD, Medical Biotechnology, Tabriz University of Medical Sciences, Iran. 2014 – 2021

MSc, Medical Biotechnology, Golestan University of Medical Sciences, Iran. 2010 – 2013

BSc, Cell & Molecular Biology, Azerbaijan University of Tarbiat Moallem, Iran. 2005 – 2009

SELECTED PUBLICATIONS (Full list: Google Scholar | ORCID: 0000-0001-6852-9865)

Biancardi F, **Keyhanvar N**, Zhurikhina A et al. Multi-organ collection from mouse models for aging studies. *JoVE*. 2025. (co-first author)

Turkalj EM, Kuljanishvili M, Kang G, Liu I, Ghent C, Waltier Soriano C, **Keyhanvar N**, et al. Cortical organoids reveal human-specific roles of METTL5 in neurodevelopment via regulation of CHCHD2. *bioRxiv*. 2025.

Keyhanvar N, Zarghami N, Seifalian A et al. The combined thermoresponsive cell-imprinted substrate, induced differentiation, and "KLC Sheet" formation. *Advanced Pharmaceutical Bulletin*. 2021.

Keyhanvar N, Zarghami N, Bleisinger N et al. Cell-based endometrial regeneration: current status and future perspectives. *Cell and Tissue Research*. 2021.

Yazdani Y, **Keyhanvar N**, Tabaraei A. Cloning and functional assessment of recombinant human hepcidin-25 in the baculovirus expression system. *Brazilian Archives of Biology and Technology*. 2014.

Yazdani Y, **Keyhanvar N**, Kalhor HR, Rezaei A. Functional analyses of recombinant mouse hepcidin-1 in cell culture and animal model. *Biotechnology Letters*. 2013.

AWARDS

Selected for oral presentation – Bakar Aging Research Institute (BARI) Symposium, San Francisco, US. 2025

Volunteer – 50th RESI Conference, J.P. Morgan Healthcare Week, San Francisco, US. 2025

Invited peer reviewer – BioImpacts journal (6 manuscripts), Iran. 2023

3rd Prize – Technology, Entrepreneurship & Research Festival (FAKT), Iran. 2016

Competitive research grant award – Stem Cell & Regenerative Medicine Institute, Iran. 2014

1st Prize, Best Oral Presentation & Best Abstract – National Congress on Biotechnology, Iran. 2012